EPSR-test-ex101 Coarse graining with distributed molecule scale factors

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the coarse-graining option of the `epsr` engine of EPSR [1,2], on when the first value of `qinscalefc` is > 1: per-molecule size factors `scalefacmol` (`.ato` molecule headers) and per-type sphere parameters `natomsph radminsph radmaxsph cgmolfac` define form factors and scale the reference-potential distances of molecule pairs [3]. This case gives the molecules different scale factors, so the scaling of the pair energies and the binning of form factors are exercised.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`qinscalefc`), `inita` (coarse-graining block: bins `assignatomscalefac`, `sphereformfactor`, `intraformfactor`, `intraformfactorself`), `gaussianqr2`
- `molfit.f`: `en_mole_pair`, `en_change`, `en_atom_inter_scale` (pair scale factor `scalefac`)
- `ato_manipulation_rev4.f`: `atogen` (scaled `potex`, `wavinter`); `ato_read_and_save.f`: `minitc`, `moloutc`

Rust (port, rust/epsr-core/src/):

- `coarsegrain.rs` (new): `State::inita_coarsegrain`, `gaussianqr2`, `atogen_accumulate`, `atogen_normalise`
- `epsr.rs`: `Epsr::setup` (`qinscalefc`); `structure.rs`: `inita`, `atogen`
- `energy.rs`: `en_mole_pair`, `en_change`, `en_atom_inter_scale`; `ato.rs`: `minitc`, `moloutc`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Coarse graining represents atoms by uniform spheres (form factors from `gaussianqr2` of the inner and outer radius); a molecule pair scales the reference-potential distance by `scalefac` = `0.5*(scalefacmol(i)+scalefacmol(j))`. Here `scalefacmol` cycles through 1.0/1.25/1.5/1.7 for DMSO and 0.8/1.0/1.2 for water, giving 4 and 3 form-factor bins with bin averaging and type minima below 1.0; `selfbroadpwr` 0.33 gives `scalefacmoluse` = (atoms per molecule)^0.33.

Inputs

Generated from EPSR-test by `scripts/make_cg.py <folder> <mode>` (edits `.EPSR.inp` keywords, the `.ato` `scalefacmol` words and type columns). Mode for this case: `.EPSR.inp` `qinscalefc 1.5 0.33`, `ntimes 3`; `.ato` scale factors as above; `.pcof` with the `rminex` values of EPSR-test-ex1 (S-S 4.0 and two water pairs), so the scaled close-contact count `potex` of `atogen` matters.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The pair scale factor changes the Monte Carlo energies: first-pass energy per molecule -30.18 kJ/mol against -39.51 kJ/mol for the same input with `qinscalefc 1.0` (README). The three form-factor files are compared byte for byte with the 27 standard outputs.

Notes

Fortran: form factors and `wavinter` are computed but never used for S(Q) (the coarse-graining code in `gensq` is commented out).

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.